Bravais lattices

The 5 two-dimensional and 14 three-dimensional lattice types

crystal systems · conventional cells · symmetry · cell properties

2D 5 lattices · 4 systems · 10 point groups · 17 wallpaper groups

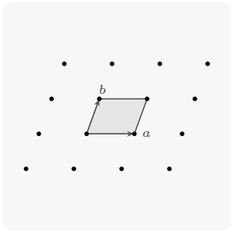
3D 14 lattices · 7 systems · 32 point groups · 230 space groups

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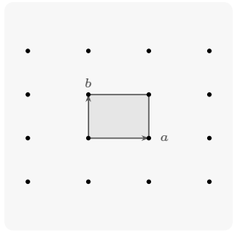
• lattice point • centring point — conventional cell edge a, b, c unit-cell vectors the cell drawn in each panel is the conventional cell

Two dimensions

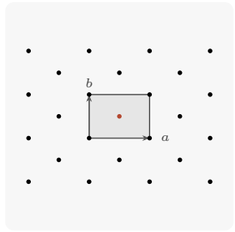
5 lattices · 4 crystal systems · 17 wallpaper groups · cell area $ab\sin\gamma$



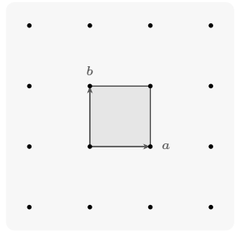
Oblique **p**
primitive, 1 point
$a \neq b, \gamma \neq 90°$
holohedry 2
$p1$, $p2$



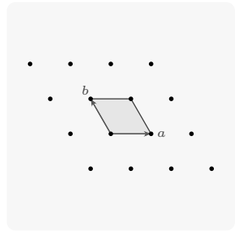
Rectangular **p**
primitive, 1 point
$a \neq b, \gamma = 90°$
holohedry $2mm$
$p1m1$, $p1g1$, $p2mm$, $p2mg$, $p2gg$



Rectangular **c**
centred, 2 points
$a \neq b, \gamma = 90°$
holohedry $2mm$
$c1m1$, $c2mm$



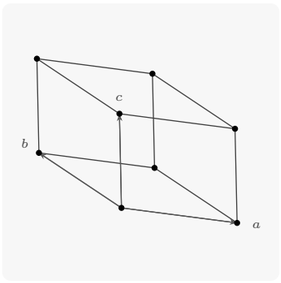
Square **p**
primitive, 1 point
$a = b, \gamma = 90°$
holohedry $4mm$
$p4$, $p4mm$, $p4gm$



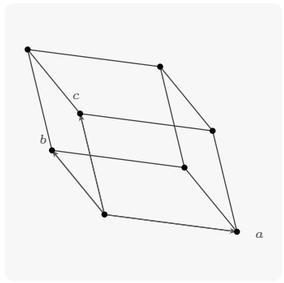
Hexagonal **p**
primitive, 1 point
$a = b, \gamma = 120°$
holohedry $6mm$
$p3$, $p3m1$, $p31m$, $p6$, $p6mm$

Three dimensions

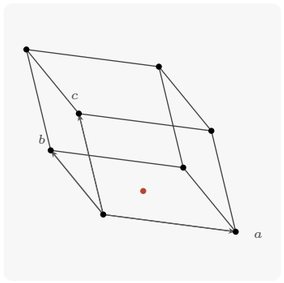
14 lattices · 7 crystal systems · 32 point groups · 230 space groups



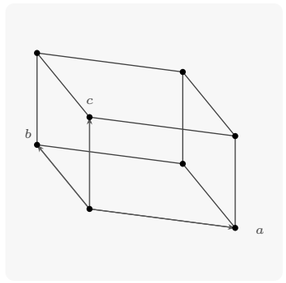
Triclinic **P**
primitive, 1 point
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$
holohedry $\bar{1}$



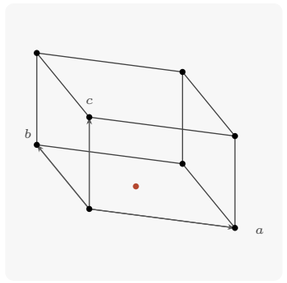
Monoclinic **P**
primitive, 1 point
$a \neq b \neq c$
$\alpha = \gamma = 90° \neq \beta$
holohedry $2/m$



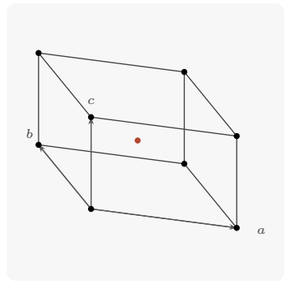
Monoclinic **C**
base-centred, 2 points
$a \neq b \neq c$
$\alpha = \gamma = 90° \neq \beta$
holohedry $2/m$



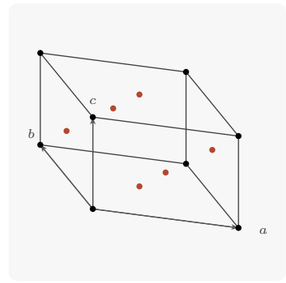
Orthorhombic **P**
primitive, 1 point
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$
holohedry mmm



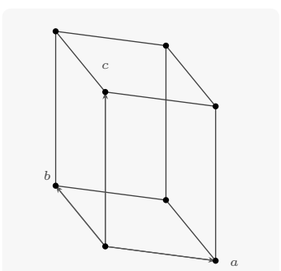
Orthorhombic **C**
base-centred, 2 points
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$
holohedry mmm



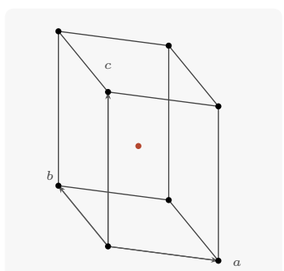
Orthorhombic **I**
body-centred, 2 points
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$
holohedry mmm



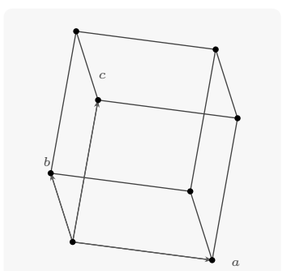
Orthorhombic **F**
face-centred, 4 points
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$
holohedry mmm



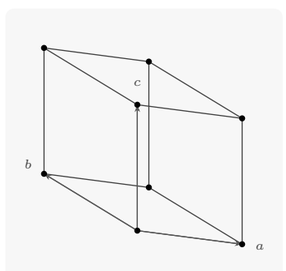
Tetragonal **P**
primitive, 1 point
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$
holohedry $4/mmm$



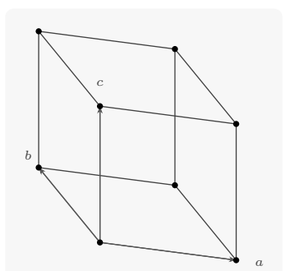
Tetragonal **I**
body-centred, 2 points
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$
holohedry $4/mmm$



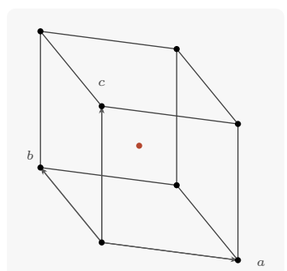
Trigonal **R**
rhombohedral, 1 point
$a = b = c$
$\alpha = \beta = \gamma \neq 90°$
holohedry $\bar{3}m$



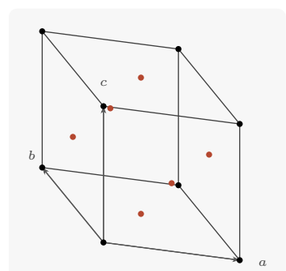
Hexagonal **P**
primitive, 1 point
$a = b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$
holohedry $6/mmm$



Cubic **P**
primitive, 1 point
$a = b = c$
$\alpha = \beta = \gamma = 90°$
holohedry $m\bar{3}m$



Cubic **I**
body-centred, 2 points
$a = b = c$
$\alpha = \beta = \gamma = 90°$
holohedry $m\bar{3}m$



Cubic **F**
face-centred, 4 points
$a = b = c$
$\alpha = \beta = \gamma = 90°$
holohedry $m\bar{3}m$

Symmetry and unit-cell properties of the seven crystal systems

point groups and space groups counted per system

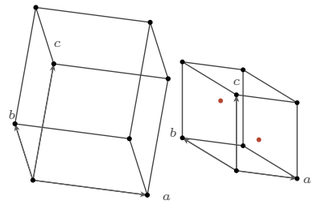
Crystal system	Lattices	Cell parameters	Holohedry	Point groups	Space groups	Volume of conventional cell	Points per cell
Triclinic	P	$a \neq b \neq c;\ \alpha \neq \beta \neq \gamma \neq 90°$	$\bar{1}$	2	2	$abc\left(1 + 2\cos\alpha\cos\beta\cos\gamma - \cos^2\alpha - \cos^2\beta - \cos^2\gamma\right)^{1/2}$	1
Monoclinic	P, C	$a \neq b \neq c;\ \alpha = \gamma = 90° \neq \beta$	$2/m$	3	13	$abc\sin\beta$	1, 2
Orthorhombic	P, C, I, F	$a \neq b \neq c;\ \alpha = \beta = \gamma = 90°$	mmm	3	59	abc	1, 2, 2, 4
Tetragonal	P, I	$a = b \neq c;\ \alpha = \beta = \gamma = 90°$	$4/mmm$	7	68	a^2c	1, 2
Trigonal	R	$a = b = c;\ \alpha = \beta = \gamma \neq 90°$	$\bar{3}m$	5	25	$a^3\left(1 - 3\cos^2\alpha + 2\cos^3\alpha\right)^{1/2}$	1 (3 in hex.)
Hexagonal	P	$a = b \neq c;\ \alpha = \beta = 90°,\ \gamma = 120°$	$6/mmm$	7	27	$\frac{\sqrt{3}}{2}\,a^2c$	1
Cubic	P, I, F	$a = b = c;\ \alpha = \beta = \gamma = 90°$	$m\bar{3}m$	5	36	a^3	1, 2, 4
Total	**14**			**32**	**230**		

Not Bravais lattices. hcp, diamond, graphene and NaCl are not lattice types: each is a Bravais lattice carrying a multi-atom basis. hcp is the hexagonal P lattice with a two-atom basis — its cell holds two atoms but contributes one lattice point. Graphene sits on one of the five two-dimensional lattices.

Centring is not always a new lattice. C-centred tetragonal is not a separate type: its primitive cell is a smaller tetragonal cell of the same symmetry. C-centred cubic is not a lattice type at all — centring a cubic cell destroys the cubic symmetry, and the result is a primitive tetragonal lattice with $c/a = \sqrt{2}$. Base-centred monoclinic has C and I settings: one lattice, two cells.

Holohedry. The point group of the lattice itself, the largest point group of the crystal system. It describes the symmetry of the infinite lattice, which a cell drawn for it need not visibly show. Point groups and space groups are counted per system, not per lattice.

Why only 14. A lattice type is fixed by the point symmetry of its cell; centring adds a point without lowering that symmetry, and further cell choices are absorbed into the primitive cell. Searching the seven systems for compatible centrings gives $1 + 2 + 4 + 2 + 1 + 1 + 3 = 14$.



The rhombohedral lattice in its two settings: the rhombohedral cell (left, one lattice point) and the hexagonal cell (right, three — one at the corners, two inside).